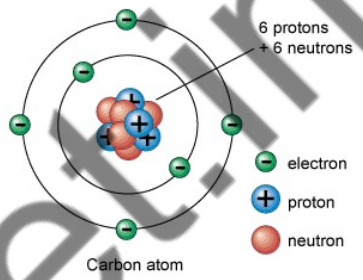
Acid Base & Salt

Introduction of Carbon:

Carbon is the 15th most abundant element in earth's crust and fourth most abundant element in universe by mass. Food, clothes, medicines, books, or many of the things that you listed are all based on this versatile element carbon. In other hand, all living structures are carbon based. Carbon plays very important role in our life.

- Symbol of Carbon = C
- Atomic number of carbon = 6
- Electronic configuration = 2, 4
- Valence electrons of carbon = 4
- Nature of carbon = Non-metal
- Isotopes = ^{12}C, ^{13}C and ^{14}C



Since, valency of carbon is equal to four, thus carbon is called tetravalent.

Four valency of carbon means, carbon reacts with four atoms of same or different elements to form compound.

Presence :

Carbon forms very large numbers of compound in nature. The earth's crust has only 0.02% carbon in the form of minerals (like carbonates, hydrogencarbonates, coal and petroleum) and the atmosphere has 0.03% of carbon dioxide. Carbon is common element which is found in universe and forms many different compounds. The many non-living and living things surrounding us made up of carbon and its compound like plants animals, sugar, fuels, papers, foods, textiles, fabrics, medicines, cosmetics etc.

Allotropes of carbon :

Allotrope : An element, in different forms, having different physical properties but similar chemical properties is known as allotropes of that element . Carbon has three well known allotropes which are **graphite**, **daimond** and **buck minster fullerene**. These are formed by carbon atoms.

Graphite :- In graphite each carbon atom is bonded to three other carbon atoms in the same plane giving a hexagonal array. One of these bonds is a double-bond, and thus the valency of carbon is satisfied. Graphite structure is formed by the hexagonal arrays being placed in layers one above the other. Graphite is also a very good conductor of electricity unlike other non-metals

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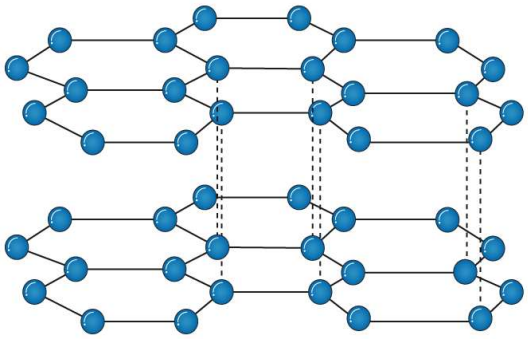
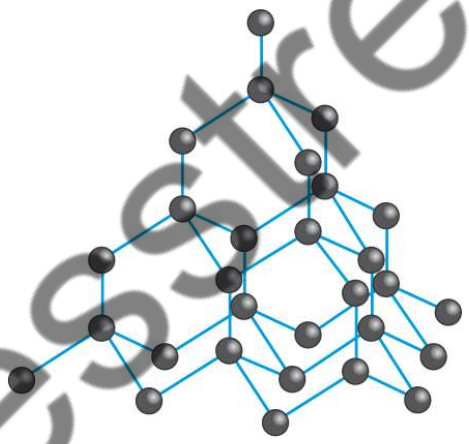


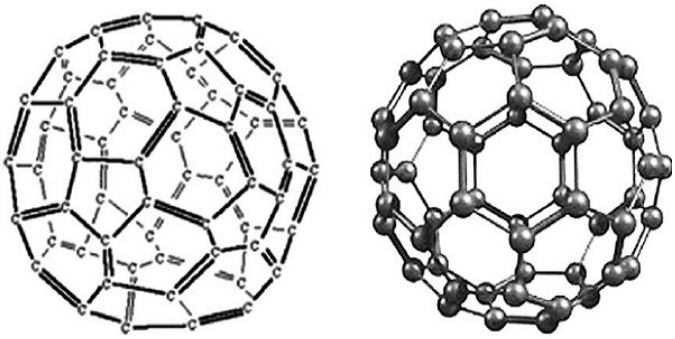
graphite

Diamond :- In diamond, each carbon atom is bonded to four other carbon atoms forming a rigid three-dimensional structure. Diamond is the hardest substance known while graphite is smooth and slippery. Diamonds can be synthesized by subjecting pure carbon to very high pressure and temperature. These synthetic diamonds are small but are otherwise indistinguishable from natural diamonds.



diamond

Fullerenes :- Fullerenes form another class of carbon allotropes. The first one to be identified was C-60 which has carbon atoms arranged in the shape of a football. Since this looked like the geodesic dome designed by the US architect Buckminster Fuller, the molecule was named fullerene.



Fullerenes

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BONDING IN CARBON:

Carbon has four electrons in its outermost shell and needs to gain or lose four electrons to attain noble gas configuration. Thus, carbon can achieve stable configuration by following two ways:

(i) **By losing four electron**:- It could gain four electrons forming C^{4-} anion. But it would be difficult for the nucleus with six protons to hold on to ten electrons, that is, four extra electrons.

(ii) **By gain of four electron** :- It could lose four electrons forming C^{4+} cation. But it would require a large amount of energy to remove four electrons leaving behind a carbon cation with six protons in its nucleus holding on to just two electrons.

Carbon overcomes this problem by sharing its valence electrons with other atoms of carbon or with atoms of other elements. Not just carbon, but many other elements form molecules by sharing electrons in this manner.

Chemical Bond : The force exerts between two atoms of elements is called chemical bond.

(i) Ionic Bond : The bond formed by transfering electrons completly is called ionic bond.

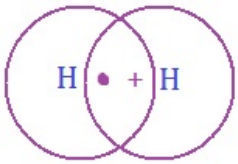
$$Na^{+} + Cl^{-} \longrightarrow NaCl$$

(ii) Covalent Bond : Bonds which are formed by the sharing of an electron pair between two atoms are known as covalent bonds.

Types of Covalent Bond : There are three types of covalent bond.

(A) Single Covalent Bond : A covalent bond formed by sharing of one electron pair between two atoms is called a single covalent bond. A single bond is represented by a horizontal dash (–) line.

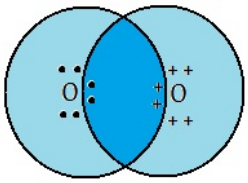
Example: H - H, Cl - Cl, Br - Br



H–H

(B) Double Covalent Bond : A covalent bond formed between two atoms by sharing of two electron pairs is called double covalent bond. A double bond is represented by simply two parallel horizontal dashed (=) lines

Example: O=O [oxygen to oxygen]



O = O

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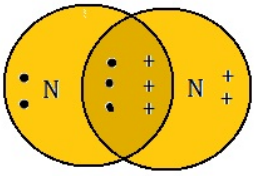
(C) Triple Covalent Bond : A covalent bond formed by sharing of three electrons between two atoms is called triple covalent bond.. A triple bond is represented by three parallel horizontal dashed (≡) lines.

Example: N ≡ N [Nitrogen to nitrogen]



N≡N

Properties of Compound formed by Covalent Bond:

(i) Covalently bonded molecules are seen to have strong bonds within the molecule.

(ii) intermolecular forces are small.

(iii) These have low melting and boiling points of these compounds.

(iv) These compounds are generally poor conductors of electricity.

Some Other Properties of Carbon:

1. Catenation : Carbon has the unique ability to form bonds with other atoms of carbon, giving rise to large molecules. This property is called catenation.

NOTE:- The nature of the covalent bond enables carbon to form a large number of compounds.

2. Tetravalency : Carbon has a valency of four, it is capable of bonding with four other atoms of carbon or atoms of some other mono-valent element. This property of carbon is known as **tetravalency** of carbon.

Some Features of Carbon Bond:

(i) Carbon forms with most other elements are very strong making these compounds exceptionally stable.

(ii) One reason for the formation of strong bonds by carbon is its small size.

(iii) This enables the nucleus to hold on to the shared pairs of electrons strongly.

Differences between bonds formed by carbon and other larger atoms:

the formation of strong bonds by carbon is its small size. This enables the nucleus to hold on to the shared pairs of electrons strongly. The bonds formed by elements having larger atoms are much weaker.

A Large number of Organic Compounds are formed by carbon:

A Large number of Organic Compounds are formed by carbon in nature due to its following features:

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(i) Forming covelent bond : Carbon makes a large number of compound in nature due to form covelent bond.

(ii) Catenation: The carbon-carbon bond is very strong and hence stable. This gives us the large number of compounds with many carbon atoms linked to each other.

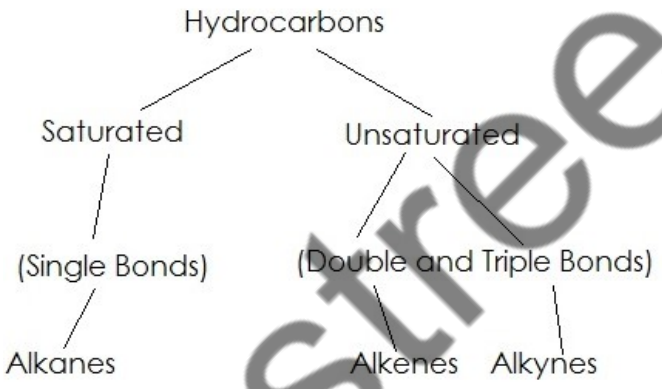
(iii) Tetravalency: Since carbon has a valency of four, it is capable of bonding with four other atoms of carbon or atoms of some other mono-valent element.

Hydrocarbon :-

Hydrogen + Carbon = Hydrocarbon

All carbon compounds which contain just carbon and hydrogen are called hydrocarbons.

Hydrocarbons are of two types: Saturated hydrocarbons and Unsaturated hydrocarbons.



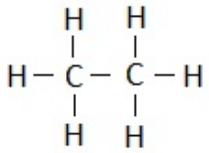
Formulae of organic compounds:

(i) General formula : General formula represents a function for **n** numbers of each atom in a molecule.
Ex : for alkane : C_nH_{2n+2}

(ii) Molecular formula : Molecular formula represents actual number of atoms of a molecule.
Ex : For Ethane: C_2H_6

(iii) Condensed formula : Condensed formula represents group of atoms linked together to each carbon atom.
Ex : For Ethane : CH_3CH_3

(iv) Structural formula : It represents exact arrangement of atom of a molecule.
Ex : For Ethane:



Ethane (C_2H_6)

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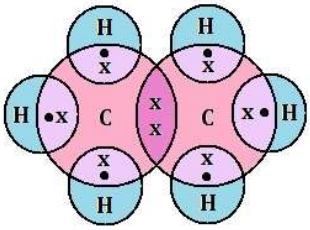
(v) Electronic formula : Electronic formula represents the sharing of electrons amongs atoms of a molecules.

Ex : For Ethane :



Saturated Carbon Compounds:

Carbon compounds that have chain of carbon atoms linked together with single covalent bonds are called saturated compounds.

Example : Methane (CH_4), Ethane (C_2H_6), chloromethane (CH_3Cl), etc.

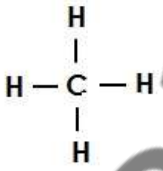
Saturated hydrocarbons are known as Alkanes.

General formula for Alkanes : $\boxed{C_nH_{2n+2}}$, Where, n = number of carbon atoms

(1) If n = 1

$\therefore$ CnH2n + 2

$= C_1H_{2 \times 1 + 2} = CH_4$

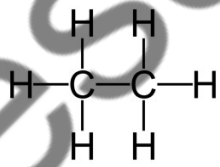


Saturated Hydrocarbon having one carbon atom is called **METHANE**

(2) If n = 2

$\therefore C_nH_{2n+2}$

$= C_2H_{2 \times 2 + 2} = C_2H_6$

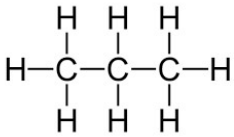


Saturated Hydrocarbon having two carbon atoms is called **ETHANE**

(3) If n = 3

$\therefore C_nH_{2n+2}$

$= C_3H_{2 \times 3 + 2} = C_3H_8$



Saturated Hydrocarbon having two carbon atoms is called **PROPANE**

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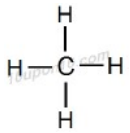
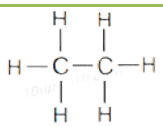
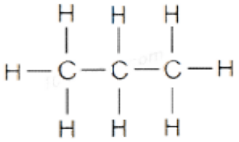
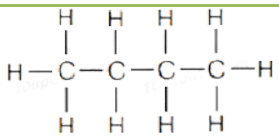
List given below for Molecular, Condensed and structural formula for Alkanes upto n=10,

Name of Alkanes	Molecular Formula	Condensed structural Formula
Methane	CH_4	CH_4
Ethane	C_2H_6	CH_3CH_3
Propane	C_3H_8	$CH_3CH_2CH_3$
Butane	C_4H_{10}	$CH_3CH_2CH_2CH_3$
Pentane	C_5H_{12}	$CH_3CH_2CH_2CH_2CH_3$
Hexane	C_6H_{14}	$CH_3CH_2CH_2CH_2CH_2CH_3$
Heptane	C_7H_{16}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$
Octane	C_8H_{18}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3$
Nonane	C_9H_{20}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$
Decane	$C_{10}H_{22}$	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$

Structural Formula of Alkane:

एल्केन का नाम	अणु सूत्र	संरचना सूत्र
Methane	CH_4	
Ethane	C_2H_6	
Propane	C_3H_8	
Butane	C_4H_{10}	

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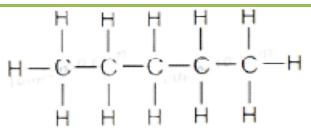
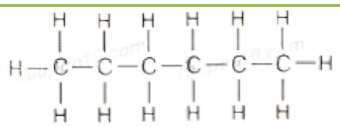
Pentane	C_5H_{12}	
Hexane	C_6H_{14}	

Unsaturated Carbon Compounds :

Hydrocarbons with double or triple bonds are called Unsaturated Hydrocarbons or Unsaturated Carbon compounds.

Example: Ethylene (C_2H_4), Acetylene (C_2H_2), etc.

- Unsaturated Hydrocarbons can be further divided into two types :
  (1) Alkene : Hydrocarbons with double bonds
  (2) Alkyne : Hydrocarbons with triple bonds.

(1) Alkene : Hydrocarbons with Double Bonds

Hydrocarbons with double bonds are called ALKENE.

Alkane – ane + ene = Alkene

General Formula of Alkene = C_nH_{2n}

(1) If n = 1

∴ CnH2n

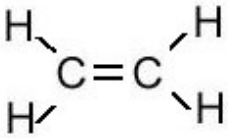
= $C_1 H_{2 \times 1} = CH_2$

This compound **does not exist.**

(2) If n = 2

∴ CnH2n

= $C_2 H_{2 \times 2} = C_2H_4$



This compound is known as **Ethene**

(3) If n = 3

∴ C_nH_{2n}

= $C_3 H_{2 \times 3} = C_3H_6$

This compound is known as **Propene**

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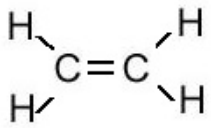
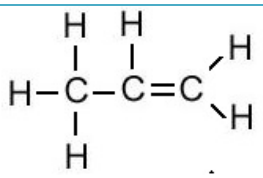
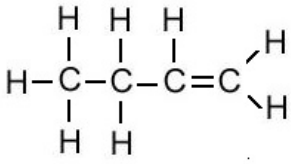
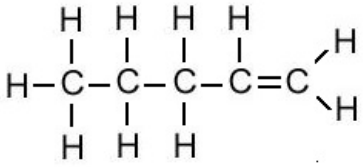
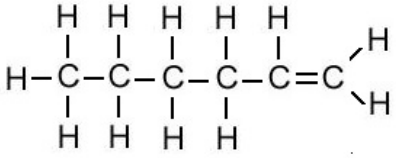
List given below for Molecular, Condensed and structural formula for Alkenes upto n=10,

Name of Alkenes	Molecular Formula	Condensed structural Formula
Ethene	C_2H_4	$CH_2=CH_2$
Propene	C_3H_6	$CH_3CH=CH_2$
Butene	C_4H_8	$CH_3CH_2CH=CH_2$
Pentene	C_5H_{10}	$CH_3CH_2CH_2CH=CH_2$
Hexene	C_6H_{12}	$CH_3CH_2CH_2CH_2CH=CH_2$
Heptene	C_7H_{14}	$CH_3CH_2CH_2CH_2CH_2CH=CH_2$
Octene	C_8H_{16}	$CH_3CH_2CH_2CH_2CH_2CH_2CH=CH_2$
Nonene	C_9H_{18}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH=CH_2$
Decene	$C_{10}H_{20}$	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH=CH_2$

Structural Formula of Alkene :

Name of Alkenes	Molecular Formula	Structural Formula
Ethene	C_2H_4	
Propene	C_3H_6	
Butene	C_4H_8	
Pentene	C_5H_{10}	
Hexene	C_6H_{12}	

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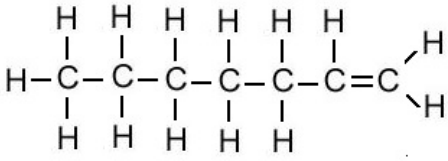
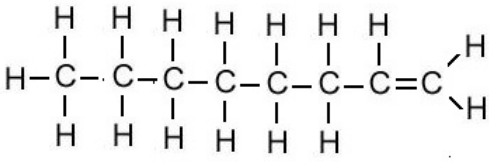
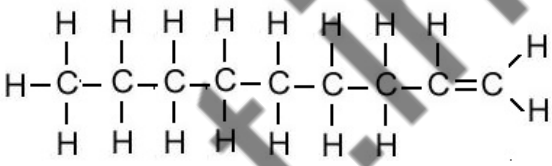
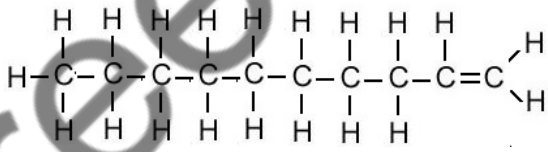
Heptene	C_7H_{14}	
Octene	C_8H_{16}	
Nonene	C_9H_{18}	
Decene	$C_{10}H_{20}$	

(2) Alkyne : Hydrocarbons with Triple bond

Unsaturated hydrocarbon with triple bond is called **Alkyne.**

Alkane – ane + yne = Alkyne

General Formula of Alkyne = $\boxed{C_nH_{2n-2}}$

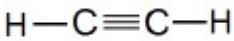
(1) If n = 1

$\therefore C_nH_{2n-2}$

$= C_1 H_{2 \times 1-2} = CH_0$

This compound **does not exist.**

(2) If n = 2



$\therefore C_nH_{2n-2}$

$= C_2 H_{2 \times 2-2} = C_2H_2$

This unsaturated hydrocarbon is called **Ethyne** [Ethane – ane + yne = Ethyne]

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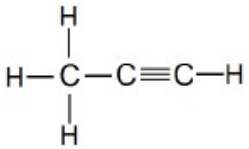
Ethyne or Acetylene

(3) If n = 3



$\therefore C_nH_{2n-2}$

$= C_3H_{2 \times 3 - 2} = C_3H_4$

This unsaturated hydrocarbon is called **Propyne** [Propane – ane + yne = Propyne]

Similarily we can find molecular formula for propyne, 1- butyne, 1- pentyne and so on .....

Name of Alkenes	Molecular Formula	Condensed structural Formula
Ethyne	C_2H_2	$CH{\equiv}CH$
Propyne	C_3H_4	$CH{\equiv}CCH_3$
1-Butyne	C_4H_6	$CH{\equiv}CCH_2CH_3$
1-Pentyne	C_5H_8	$CH{\equiv}CCH_2CH_2CH_3$
1-Hexyne	C_6H_{10}	$CH{\equiv}CCH_2CH_2CH_2CH_3$
1-Heptyne	C_7H_{12}	$CH{\equiv}CCH_2CH_2CH_2CH_2CH_3$
1-Octyne	C_8H_{14}	$CH{\equiv}CCH_2CH_2CH_2CH_2CH_2CH_3$
1-Nonyne	C_9H_{16}	$CH{\equiv}CCH_2CH_2CH_2CH_2CH_2CH_2CH_3$
1-Decyne	$C_{10}H_{18}$	$CH{\equiv}CCH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$

Defferences between saturated and unsaturated carbon compound :

Saturated Compound	Unsaturated Compound
1. It has single bond between carbon atoms. 2. There occurs substitution reaction in it. 3. Example : Alkanes. 4. They are less reactive than the unsaturated carbon compound.	1. It has double and triple bond between carbon atoms. 2. There occurs addition reaction in it. 3. Example : Alkenes, Alkynes. 4. They are more reactive than the saturated carbon compounds.

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Skeletons of Carbon Atoms:

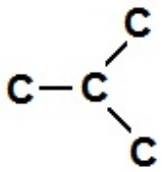
(i) Straight Carbon chain : When carbon atoms are linked together in straight chain.

Ex :-

C – C – C - C

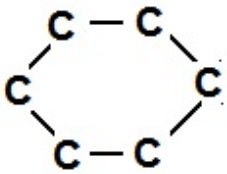
(ii) Branches Carbon chain : When carbon atoms are linked together in straight chain.

Ex :-



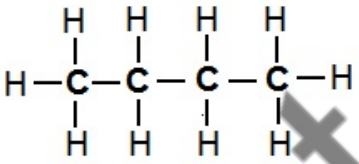
(iii) Ring Carbon chain : When carbon atoms are linked together in ring shape.

Ex :-

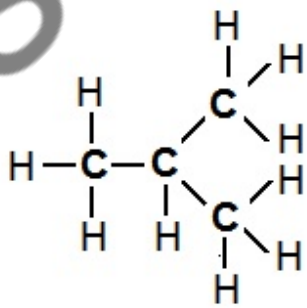


Structural Isomers :- Hydrocarbons having same molecular formula but different structural formula are called Structural Isomers.

Example 1: Structural Isomers of Butane; Whoose molecular formula is C_4H_{10}.

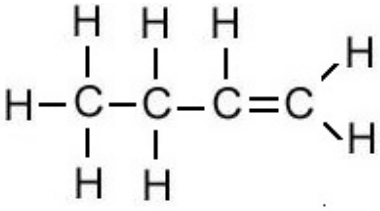


Structure Isomer (I) of Butane

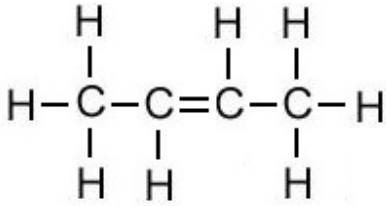


Structure Isomer (II) of Butane

Example 2 : Structural Isomers of Butene (double bond); Whoose molecular formula is C_4H_8.



Structure Isomer (I) of Butene



Structure Isomer (II) of Butene

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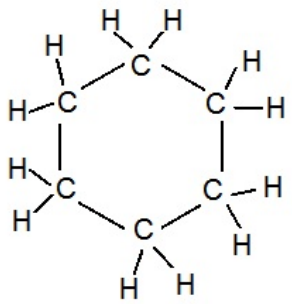
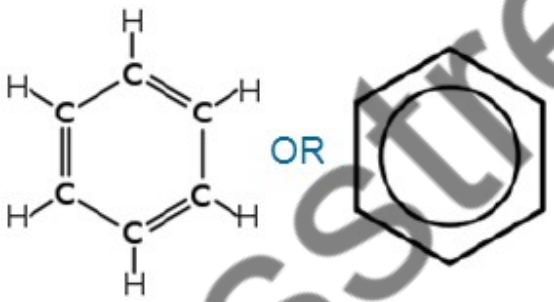
Cyclic Hydrocarbon (Hydrocarbon having ring structures):-

When carbon atoms form chain in the form of ring, such hydrocarbon is called cyclic hydrocarbon.

Ex :- Structure of cyclohexane molecule (C_6H_{12})



Ex :- Structure of Benzene Molecules (C_6H_6)



Functional Group :

Functional group is a group of an atom or atoms in any carbonic compound which are bonded each other in special manner. That is generally region of chemical reactivity in carbonic atoms. Oxygen, chlorine, sulpher, nitrogen and other elements can be presence as a part of a functional group in carbonic compounds.

Hetroatoms : The element which replaces hydrogen in a compound is called hetroatom.

Ex : Oxygen, chlorine, sulpher, nitrogen and other elements can be presence as a part of a functional group in carbonic compounds, such elements are called hetroatoms.

Some List of Functional Groups :

(i) Halogen : Halogens are non-metal elements like fluorine, chlorine, bromine and Iodine etc situated in modern periodic table in group 17.

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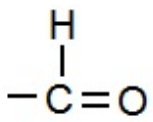
Functional Group	formula of functional group	Hetroatom
Halogen	- Cl (Sufix is used Chloro) - Br (Sufix is used Bromo) - I (Sufix is used Iodo)	Cl (Chlorine) Br (Bromine) I (Iodine)

(ii) Alcohol : Alcohol is another function group which joining to a series of hydrocarbon forms the group of atoms

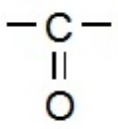
Example : - OH

(-OH) joins to hydrocarbon like alkane form various type of alcohols, such methanol, ethanol and propanol etc.

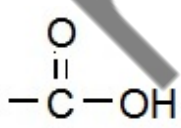
(iii) Aldehyde : It is a functional group in which carbon atom linked with a single oxygen atom in double bond along with one hydrogen atom.



(iv) Ketone : Ketone is also a functional group which joins to hydrocarbons form a series of molecules. In ketone carbon atom linked with a single oxygen atom in double bond.



(v) Carboxylic acid: It is also a functional group, in which a carbon atom is linked with oxygen atom in double bond and also linked with hydroxide.



Homologous Series:

A series of compounds in which the same functional group substitutes for hydrogen in a carbon chain is called a homologous series.

Ex:-

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Carbon chain with alkane	Homologous Series with halogen (-Cl)	Homologous Series with halogen (-Br)	Homologous Series with halogen (-I)	Homologous Series with Alcohol (-OH)	Homologous Series with Aldehyde (-CHO)
CH_4	CH_3-Cl	CH_3-Br	CH_3-I	CH_3-OH	H-CHO
C_2H_6	C_2H_5-Cl	C_2H_5-Br	C_2H_5-I	C_2H_5-OH	CH_3-CHO
C_3H_8	C_3H_7-Cl	C_3H_7-Br	C_3H_7-I	C_3H_7-OH	C_2H_5-CHO
C_4H_{10}	C_4H_9-Cl	C_4H_9-Br	C_4H_9-I	C_4H_9-OH	C_3H_7-CHO
C_5H_{12}	C_5H_{11}-Cl	C_5H_{11}-Br	C_5H_{11}-I	C_5H_{11}-OH	C_4H_9-CHO

Properties of Homologous Series :-

- Each members of a homologous series differs from its successive members by – CH_2.
- Each members of a homologous series differs by atomic mass of 14 u

Nomenclature of carbon Compounds : Naming of organic compound systematically called as nomenclature.

IUPAC Name : International Union of Pure and Applied Chemistry (IUPAC) made some rules to name the organic compounds so that each of the organic compounds can be differentiated by their names. Such names are commonly known as IUPAC names. And process to name is called the Nomenclature of Organic Compounds.

The systematic IUPAC name of an organic compound consists of five parts.

1. **2° Prefix**
2. **1° Prefix**
3. **Root word**
4. **1° Suffix**
5. **2° Suffix**

2° Prefix + 1° Prefix + Root word + 1° Suffix + 2° Suffix

1. **2° Prefix :-** The prefix is used to indicate the side chains, substituents and low priority functional groups (which are considered as substituents).

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Ex :-

Side chain or Substituent	2^0 Prefix
$-CH_3$	methyl-
- Cl	Chloro
-F	Floro
-Br	Bromo
-I	Iodo
$-CH_2CH_3$ (or) $-C_2H_5$	ethyl-
$-CH_2CH_2CH_3$	propyl-
$-CH_2CH_2CH_2CH_3$	butyl
$-NO_2$	-nitro

2. 1° Prefix :- The 1° Prefix, like cyclo, spiro, bicyclo, are added between the 2° prefix and root word in the IUPAC name to indicate the nature of parent chain.

- The "Cyclo" Prefix is used to indicate the cyclic nature of the parent chain.
- The "Spiro" Prefix is used to indicate the spiro compound.
- The "Bicyclo" Prefix is used to indicate the bicyclic nature of the parent chain.

3. Root word :- The Word root of IUPAC name indicates the number of carbon atoms in the longest possible continuous carbon chain also known as **parent chain** chosen by a set of rules.

Number of carbon atoms in the parent chain	Root word
1	Meth
2	Eth
3	Prop
4	But
5	Pent
6	Hex
7	Hept
8	Oct
9	Non
10	Dec

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4. 1° Suffix : - It is used to indicate the degree of saturation (C-C) or unsaturation (C=C, C≡C) in the main chain. It is added immediately after the word root of IUPAC name.

Type of carbon bond	1° Suffix
C-C	-ane
C=C	-ene
C≡C	-yne

5. 2° suffix :- It is used to indicate the main functional group in the organic compound and is added immediately after the 1° suffix in the IUPAC name.

Name of Functional group	Representation	2°Suffix
carboxylic acid	-COOH	-oic acid
Ester	-COOR	alkyl-oate
Acid amide	$-CONH_2$	-amide
Nitrile	-CN	-nitrile
Aldehyde	-CHO	-al
Ketone	-CO-	-one
Alcohol	-OH	-ol
Amine	$-NH_2$	-amine
Alkene	C=C	-ene
Alkyne	C≡C	-yne

NOTE : - Note: If there are two or more functional groups in a compound, the functional group with higher priority is to be selected as main functional group, which must be indicated by a secondary suffix. The remaining functional groups with lower priority are treated as substituents and are indicated by prefixes.

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COMPLETE IUPAC NAME FORMAT AND TRICK

The systematic IUPAC name of an organic compound consists of five parts.

→ **Start numbering from left to right**

2° Prefix + 1° Prefix + Root word + 1° Suffix + 2° Suffix

It tells about details of substituents.

Substituents	2o Prefix
-Cl	chloro
- F	Floro
-Br	Bromo
-I	Iodo
-CH3	Methyl
-C2H5	Ethyl

Nature of C - Chain Cyclo / BiCyclo / Spiro

Nature of C- Chain	1o Prefix
cyclic	Cyclo
spiro	Spiro
bicyclic	Bicyclo

Number of C - atoms in principal Longest Chain

No of C - atoms in parent chain	Root word
1	Meth
2	Eth
3	Prop
4	But
5	Pent
6	Hex
7	Hept
8	Oct
9	Non
10	Dec

Nature of C-C bond

C - bonds	1o suffix
C-C bonds	-ane
C=C bonds	-ene
C≡C bonds	-yne

Details of functional group

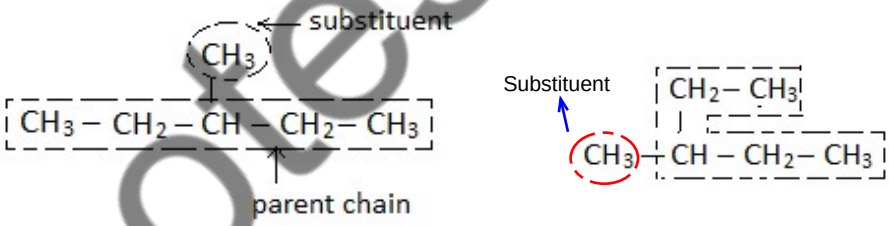
F . G	Represent
carboxylic acid	- COOH
Ester	- COOR
Acid halide	- COX
Acid amide	- CONH2
Nitrile	- CN
Aldehyde	- CHO
Ketone	- CO -
Alcohol	- OH
Amine	- NH2
Imine	= NH

RULES OF IUPAC NOMENCLATURE

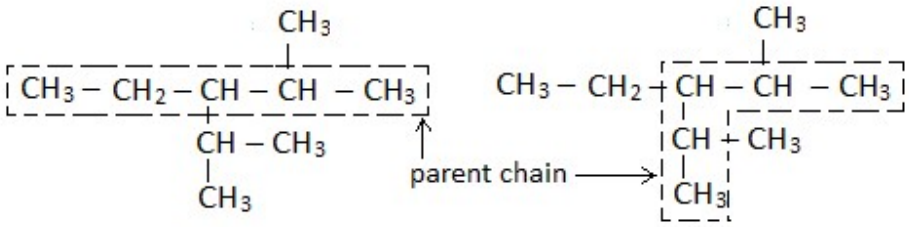
TYPE - I Compounds containing carbon to carbon single bonds and substituents only.

Rule – 1 : Selection of parent chain : The longest continuous carbon chain is selected as the parent chain.

Eg.



If two or more equally long chains are present, the chain with maximum number of substituent is selected as the parent chain.



No. of substituents = 2
WRONG selection of C- chain

No. of substituents = 3
CORRECT selection of C- chain

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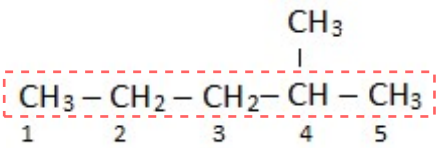
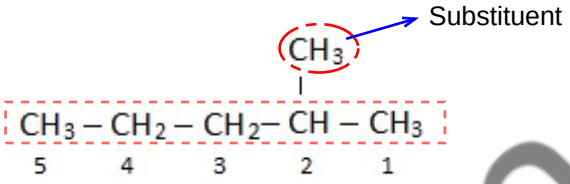
Rule - 2 : Numbering the parent chain :

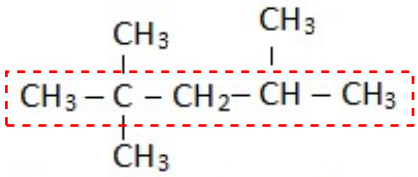
Lowest locant rule : Carbon bearing the substituent gets the lowest possible locant. Locant is a number that locate the position of substituent.



substituent gets locant 4
(WRONG numbering)

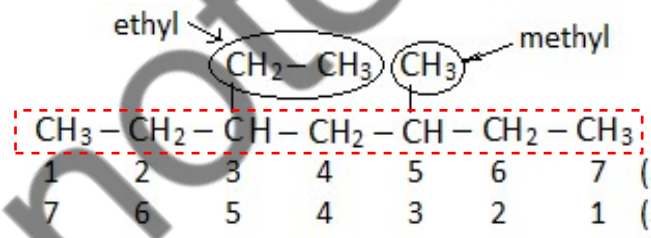


Substituent gets locant 2
(CORRECT numbering)

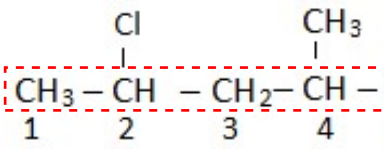


1	2	3	4	5	Set of locants : (2,2,4) (CORRECT numbering)
5	4	3	2	1	Set of locants : (2,4,4) (WRONG numbering)

Note : Here the first locant is same (i.e 2) so compare second locant which is 2 and 4 in (2,2,4) and (2,4,4). Hence (2,2,4) is lowest set of numbering and is correct numbering.
If substituents are present at equivalent position, follow alphabetical order.



1	2	3	4	5	6	7	(3,5) (correct numbering)
7	6	5	4	3	2	1	(3,5) (wrong numbering)



1	2	3	4	5	(2,4) (correct)
5	4	3	2	1	(2,4) (Wrong)

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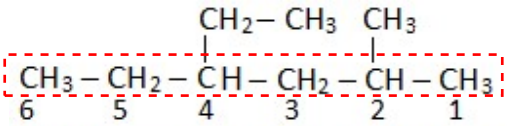
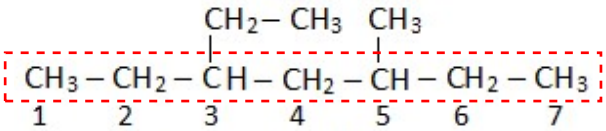
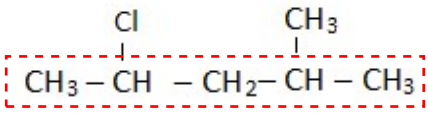
Rule - 3 – Naming : Alphabetically i.e. if more than one substituents are present then they are written in the alphabetical order.



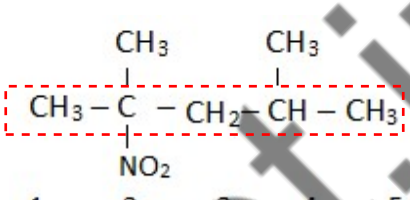
IUPAC name : 4-ethyl-2-methylhexane



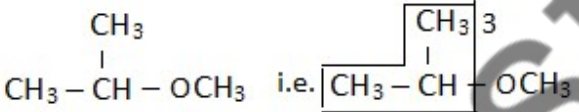
IUPAC name : 3-ethyl-5-methlyheptan



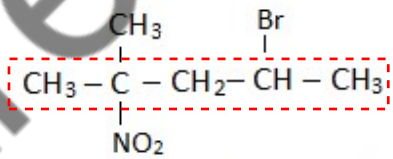
(2-chloro-4-methylpentane)



(2,4-dimethyl-2-nitropentane)



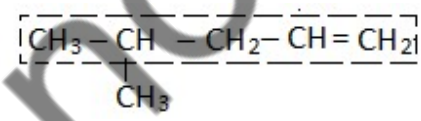
(2-methoxypropane)

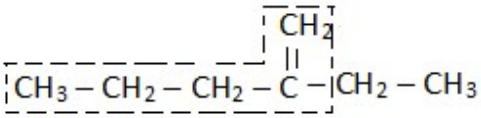


(4-bromo-2-methyl-2-nitropentane)

TYPE – II Compounds containing multiple bonds (double/ triple bonds).

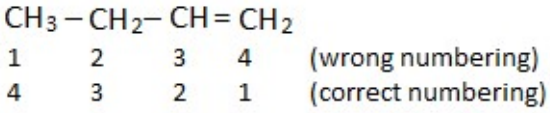
Step -1 : Selection of parent chain : The longest continuous carbon chain including multiple bond is selected as the parent chain.





Step - 2 : Numbering the parent chain :

Lowest locant rule : Carbon bearing the multiple bond gets the lowest possible locant.



If both substituent and multiple bond are present, the priority order is :

Multiple bond > Substituent

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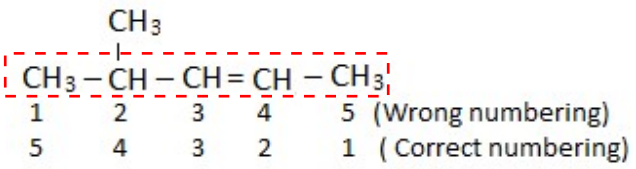
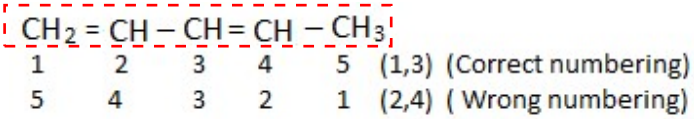
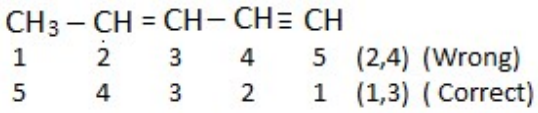
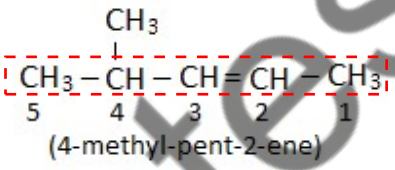




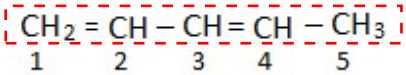


Step- 3 Naming : Alphabetically i.e. if more than one substituents and multiple bonds are present then they are written in the alphabetical order.

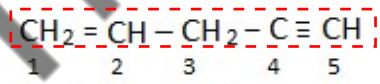
2° Prefix + 1° Prefix + Root word + 1° Suffix + 2° Suffix



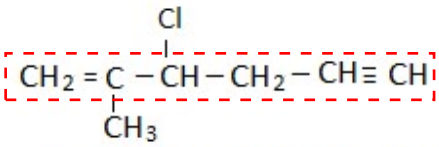
(4-methyl-pent-2-ene)



(penta-1,3-diene)



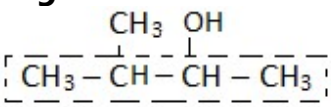
(pent-1-en-4-yne)



(3-chloro-2-methyl-hex-1-en-5-yne)

TYPE – III IUPAC Nomenclature of organic compounds containing one functional group .

Step-1 : Selection of parent chain : The longest continuous carbon chain including functional group is selected as the parent chain**. Eg.**



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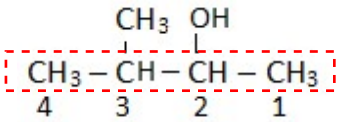
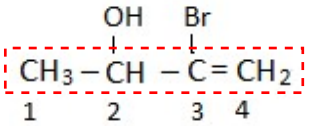
Step-2 : Numbering the parent chain : Now the final priority order for numbering is :

Functional group > Multiple bond > Substituent

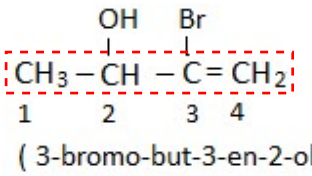




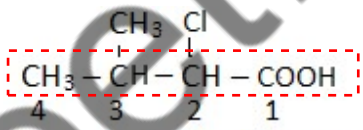
Step -3 – Naming : Always follow this format:

2° Prefix + 1° Prefix + Root word + 1° Suffix + 2° Suffix

For example,



(3-bromo-but-3-en-2-ol)



(2-chloro-3-methylbutanoic acid)

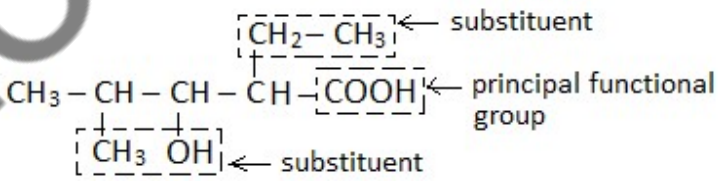
TYPE – IV IUPAC Nomenclature of organic compounds containing more than one functional groups .

The priority order of functional groups is :

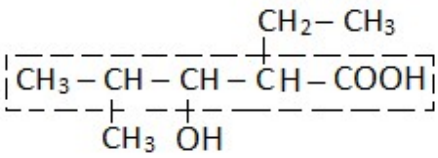
-COOH > -SO_3H > – COO- > -COX > -$CONH_2$ > -CN > -CHO > -CO- > -OH > -NH_2

During nomenclature of polyfunctional compounds, fuctional group of higher priority is taken as **principal functional** group and other functional groups are considered as **substituents.**

At first, principal functional group is identified according to priority order. **Eg**.



Step-I : Selection of parent chain : The longest continuous carbon chain with principal functional group is selected as the parent chain. **Eg.**



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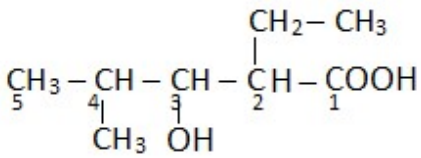
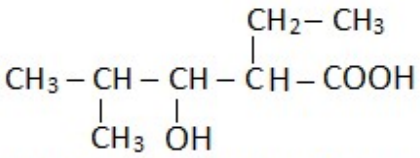
Step-II : Numbering the parent chain : Principal functional group gets lowest locant (number).



Step III – Naming : Substituents are written in alphabetical order as mentioned earlier. **Eg.**



(2-ethyl-3-hydroxy-4-methylpentanoic acid)

Chemical Properties of Carbon Compounds :-

Combustion : Combustion is the burning of compounds in air to give CO_2 and water.

(i) Combustion of Methane (CH_4) in air the reaction is as follows :

$$\mathbf{CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{heat and light}}$$

(ii) Combustion of Ethanol (CH_3CH_2OH) in air gives CO_2 water, heat and light.

$$\mathbf{CH_3CH_2OH + 3O_2 \rightarrow 2CO_2 + 3H_2O + \text{heat and light.}}$$

Above example you have seen that how carbon compounds release heat and light on burning. So, **Carbon Compounds use as fuels**.

Oxidation: Oxidation is a reaction in which carbon compounds take up oxygen in the presence of oxidising agents to form another carbon compound.

When ethyl alcohol (ethanol) is heated with alkaline potassium permanganate ($KMnO_4$) or acidified potassium dichromate ($K2Cr_2O_7$) it oxidized and gives Ethanoic acid [CH_3COOH (Acetic acid)].

$$CH_3CH_2OH + 2[O] \xrightarrow[\text{Or acidified } K_2Cr_2O_7 + \text{Heat}]{\text{alkaline } KMnO_4 + \text{Heat}} CH_3COOH + H_2O$$

Oxygen from oxidising agent $KMnO_4$ or $K_2Cr_2O_7$

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Addition Reaction :- An atom or group of atoms are added to an unsaturated compound.

"The reaction in which adds the substances is known as Addition reaction."

This reaction is commonly used in the hydrogenation of vegetable oils using a nickel catalyst.

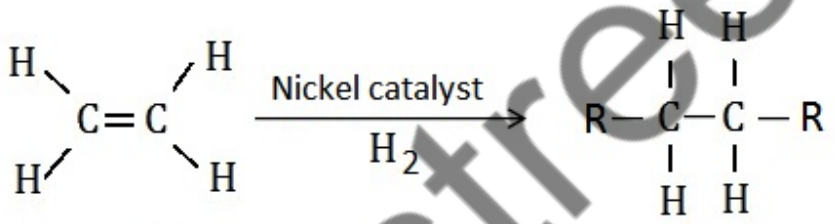
Catalysts :- Catalysts are substances that cause a reaction to occur or proceed at a different rate without the reaction itself being affected.

Unsaturated hydrocarbons add hydrogen in the presence of catalysts such as palladium or nickel to give saturated hydrocarbons.

Hydrogenation Reaction :-

Unsaturated hydrocarbons add hydrogen in the presence of catalysts such as palladium or nickel to give saturated hydrocarbons is called hydrogenation. This reaction is commonly used in the hydrogenation of vegetable oils using a nickel catalyst in industry.

Vegetable oils generally have long unsaturated carbon chains while animal fats have saturated carbon chains.



Unsaturated Hydrocarbon Saturated Hydrocarbon

Substitution Reaction :- An atom or group of atom or a group present in a saturated compound is replaced by another atom or group.

Chlorine is a hetroatom which removes hydrogen from carbon compound.

Example of substitution reaction :

When chlorine is added to hydrocarbons in the presence of sunlight It replaces hydrogen atoms one by one. This is very fast reaction. Chlorine forms functional group of halogen.

$$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl \text{ (in the presence of sunlight)}$$

Some Important Carbon Compounds :-

Ethanol (CH_3CH_2OH) : Ethanol is commonly called alcohol.

Physical Properties of Ethanol :-

(i) Ethanol is a liquid at room temperature.

(ii) It is a good solvent.

(iii) Ethanol is also soluble in water in all proportions.

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(iv) It has highly flammability.

Chemical properties of Ethanol:

(i) **Combustion :** Ethanol burns in oxygen to produce carbon dioxide and water.

(ii) **Dehydration :** Dehydration of ethanol is done by heating with concentrated sulfuric acid, which behave as a dehydrating agent.

(iii) **Oxidation :** oxidation can be made by using oxidising agent like acidified potassium dichromate or alkaline potassium permanganate.

(iv) **Estrification :** Ethanol can be reacted with carboxylic acid to form esters.

Uses of Ethanol :

(i) It is the active ingredient of all alcoholic drinks.

(ii) It is also used in medicines such as tincture iodine, cough syrups, and many tonics.

(iii) For synthesis of industrial methylated spirits.

(iv) Ethanol burns to give carbon dioxide and water and can be used as a fuel.

Harmful effects of drinking Alcohols/Ethanol :

(i) Consumption of small quantities of dilute ethanol causes drunkenness.

(ii) Short-term use of alcohol causes Slurred speech, drowsiness, vomiting and headache etc.

(iii) Long-term consumption of alcohol leads to many health problems such as alcohol poisioning, liver disease, nerve damage and permanent damage of brain etc.

Ethanoic Acid (CH_3COOH) :

Ethanoic acid is commonly called acetic acid and belongs to a group of acids called carboxylic acids.

- 5-8% solution of acetic acid in water is called **vinegar** and is used widely as a **preservative in pickles**.
- The melting point of pure ethanoic acid is 290 K (17 ^{o}C) and hence it often freezes during winter in cold climates. This gave rise to its name **glacial acetic acid**.

Properties:

(i) It has acidic nature.

(ii) Ethanoic acid is an odorless and highly smelling liquid.

(iii) Its melting point is 290 K.

Use of Ethanoic Acid/Acetic Acid :- (Carboxylic Acid Group) :-

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The uses of ethanoic acid are followings:

(i) It is used as preservative for preparing pickles in the form of vinegar.

(ii) It is used as laboratory's agent.

(iii) Manufacturing of white led.

(iv) It is used in the manufactering of Reyon fibres.

(v) Acetic acid is used as coagulant in the manufacture of rubber.

(vi) It is used as a solvent.

Ester :- The reaction by ethanol and ethanoic acid resulting forms the compound that is called Ester. The molecular formula of ester is $CH_3COOCH_2CH_3$.

Uses of Esters : -

Esters are sweet-smelling substances, the uses of esters are following.

(i) These are used in making perfumes and as flavouring agents.

(ii) These are also used to make surfactants such as soap and detergents.

(iii) Some esters can be made into polymers called polyesters.

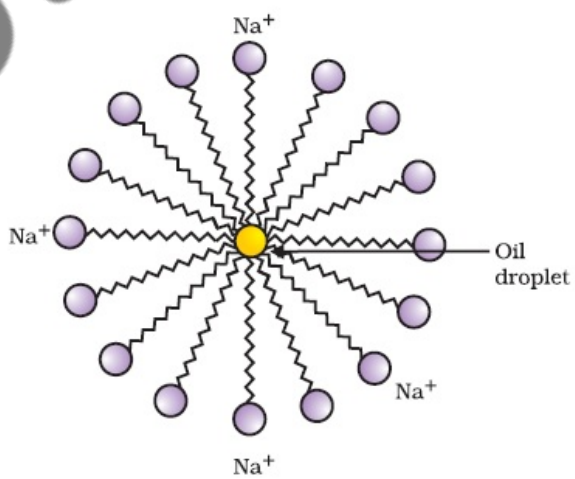
Soap And Detergents :-

Soaps :- The molecules of soap are sodium or potassium salts of long-chain carboxylic acids. The ionic-end of soap dissolves in water while the carbon chain dissolves in oil. The soap molecules, thus form structures called micelles.

Dirt in clothes contains oil which gets stuck with clothes. When cloth is dip in soap water, soap micelle helps in dissolving dirt present in cloths and water which can be washed out while rinsing, this cleans the cloths.

Micelles

Soap molecules have two ends, one end is ionic end and other one is carbon chain. Ionic end dissolves in water while carbon chain dissolves in oil. Ionic end which dissolves in water is called **hydrophilic** and carbon chain is **hydrophobic.**



The Structure of Micelle

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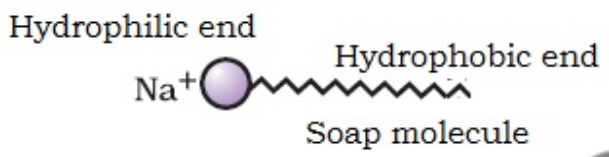
When soap is at the surface of water, inside water these molecules have a unique orientation that keeps the hydrocarbon portion out of the water. This is achieved by forming clusters of molecules in which the hydrophobic tails are in the interior of the cluster and the ionic ends are on the surface of the cluster. This formation is called a micelle.

The Micelle has two ends:

(i) Hydrophilic end : The end which dissolves in water is called hydrophilic end.

(ii) Hydrophobic end : The end which dissolves in hydrocarbons (oily substances) is called hydrophobic end.



Differences between hydrophilic end and hydrophobic end :

Hydrophilic end :

(i) It dissolves in water.

(ii) This is ionic end.

(iii) The ionic ends are on the surface of the cluster.

Hydrophobic end :

(i) It dissolves in hydrocarbons.

(ii) This is not ionic end.

(iii) The hydrophobic tails are in the interior of the cluster.

The cleaning process of soap : -

The cleaning process is characterised by soap micelle. Soap in the formation of micelles traps dirt in the centre. The micelles stay in the solution as colloid because of ion-ion repulsion do not come together to precipitate. Thus, dirt suspended in micelles easily rinsed away.

The soap micelles are large enough to scatter light. Hence a soap solution appears cloudy.

The soap does not form foam with hard water : When we wash hand with soap in hard water, we see that foam formed with very difficulty and an insoluble substance (scum) remains after washing with water.

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Detergent :-

Detergents are generally ammonium or sulphonate salts of long chain carboxylic acids. Detergents clean clothes even in hard water.

When detergent is mixed in water, the charged ends of detergent molecules do not form insoluble precipitates with the calcium and magnesium ions present in hard water and clean clothes in better way. Because of these properties detergent is preferred at the place of soap for cleaning.

Differences between soap and detergent:

Soap:

(i) The molecules of soap are sodium or potassium salts of long-chain carboxylic acids.
(ii) It does not form foam in hard water.
(iii) It increases the hardeness of water.
(iv) It forms micelle during cleaning process.

Detergent:

(i) Detergents are generally ammonium or sulphonate salts of long chain carboxylic acids.
(ii) It form foam even in hard water.
(iii) It increases the softness of water.
(iv) It does not form micelle during cleaning process.

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